

Linear Combinations Unit 2

RV University · 2nd Semester Computer Science · CIE 1 Preparation

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Chapter 1: Homogeneous Equations

1.1 Definition and General Form

A system of linear equations is homogeneous if every constant term is zero. The general matrix form is:

$$A \cdot X = 0$$

where A is the $m \times n$ coefficient matrix, X is the $n \times 1$ vector of unknowns, and 0 is the $m \times 1$ zero vector.

For a 3×3 system this expands to:

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

KEY FACT: A homogeneous system is ALWAYS consistent. $x = 0$ (the trivial solution) satisfies $Ax = 0$ for every matrix A . Therefore, an inconsistent homogeneous system cannot exist.

1.2 Solution Types

Condition	Determinant	Solution Type	Number of Solutions
$\det(A) \neq 0$	Non-zero	Trivial only	Exactly 1 ($x=0$)
$\det(A) = 0$	Zero	Non-trivial exist	Infinitely many
$\text{Rank}(A) = n$	—	Trivial only	Exactly 1
$\text{Rank}(A) < n$	—	Free variables exist	Infinitely many

Trivial Solution: The solution $X = 0$ (all variables equal zero). Always exists for any homogeneous system.

Non-trivial Solution: Any solution $X \neq 0$. Exists if and only if $\det(A) = 0$, equivalently $\text{rank}(A) < n$.

1.3 Solution Procedure

Step 1 — Determinant Test (Square Systems)

1. Form the coefficient matrix A .
2. Compute $\det(A)$.
3. If $\det(A) \neq 0 \rightarrow$ only trivial solution. Stop.
4. If $\det(A) = 0 \rightarrow$ infinitely many solutions. Proceed to Step 2.

Step 2 — Gaussian Elimination

5. Write the augmented matrix $[A \mid 0]$.
6. Apply row operations to reach Row Echelon Form (REF).
7. Count pivots: $r = \text{rank}(A)$. Free variables = $n - r$.
8. Assign parameters to free variables, back-substitute to find general solution.

Step 3 — General Solution Structure

The solution set of $Ax = 0$ is a vector space (the null space of A). Every solution can be written as a linear combination of basis vectors of the null space.

$$X = t_1 v_1 + t_2 v_2 + \cdots + t_{n-r} v_{n-r}, \quad t_i \in \mathbb{R}$$

1.4 Worked Examples

Example 1 — Trivial Solution Only

System:

$$x + y + z = 0, \quad 2x + 3y + 4z = 0, \quad 3x + 5y + 6z = 0$$

Form A and compute $\det(A)$:

$$\det(A) = 1(18-20) - 1(12-12) + 1(10-9) = -2 - 0 + 1 = -1 \neq 0$$

Result: Only trivial solution: $x = y = z = 0$.

Example 2 — Infinitely Many Solutions (2 free variables)

System:

$$x + y + z = 0, \quad 2x + 2y + 2z = 0, \quad 3x + 3y + 3z = 0$$

Augmented matrix and row reduction:

$$\begin{bmatrix} 1 & 1 & 1 & | & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 & 2 & | & 0 \end{bmatrix} \rightarrow R_2 - 2R_1, R_3 - 3R_1 \rightarrow \begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 3 & 3 & | & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Rank $r = 1$. Free variables = $n - r = 3 - 1 = 2$. Let $y = s, z = t$.

From row 1: $x = -s - t$.

$$(x, y, z) = s(-1, 1, 0) + t(-1, 0, 1), \quad s, t \in \mathbb{R}$$

Null space basis: $\{(-1, 1, 0), (-1, 0, 1)\}$ — $\dim(N(A)) = 2$

Example 3 — One Free Variable

System:

$$2x + y - z = 0, \quad 4x + 2y - 2z = 0, \quad x + y + z = 0$$

Row reduce (swap $R_1 \leftrightarrow R_3$ first for cleaner arithmetic):

$$\begin{bmatrix} 1 & 1 & 1 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & 0 \end{bmatrix} \text{ (after } R_2 - 4R_1, \text{ then } -\frac{1}{2}R_2)$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix} \text{ (} R_3 + R_2)$$

Rank $r = 2$. Free variable: $z = t$.

From row 2: $y = -3t$. From row 1: $x = 2t$.

$$(x, y, z) = t(2, -3, 1), \quad t \in \mathbb{R}$$

Null space basis: $\{(2, -3, 1)\}$ — $\dim(N(A)) = 1$

1.5 Key Formulas — Homogeneous Systems

Formula	Meaning
$\text{rank}(A) + \text{nullity}(A) = n$	Rank-Nullity (n = number of columns)
$\text{nullity}(A) = n - \text{rank}(A)$	Number of free variables = dimension of null space
$\det(A) \neq 0 \Leftrightarrow$ trivial solution only	For square $n \times n$ matrix
$\det(A) = 0 \Leftrightarrow$ non-trivial solutions exist	For square $n \times n$ matrix
Null space $N(A)$ is a subspace of $\mathbb{R}^n$	Solution set is always a vector space

Chapter 2: Linear Combinations

2.1 Definition

Linear Combination: Let V be a vector space over field F and let $v_1, v_2, \dots, v_n \in V$. A vector $v \in V$ is a linear combination of $v_1, \dots, v_n$ if there exist scalars $a_1, \dots, a_n \in F$ such that:

$$v = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

The scalars $a_1, a_2, \dots, a_n$ are called the coefficients of the linear combination.

2.2 Geometric Interpretation

In $\mathbb{R}^2$

Configuration	Result
u and v are collinear (parallel)	$au + bv$ spans a LINE through the origin
u and v are non-collinear	$au + bv$ spans the entire PLANE $\mathbb{R}^2$

In $\mathbb{R}^3$

Number of Vectors	Independence	Span
1 vector	Non-zero	Line through origin
2 vectors	Independent	Plane through origin
3 vectors	Independent	All of $\mathbb{R}^3$
Any set	Dependent	Lower-dimensional subspace

2.3 Connection to Systems of Linear Equations

Determining whether v is a linear combination of $v_1, \dots, v_n$ is equivalent to testing whether the system $Ax = v$ is consistent, where:

$$A = [v_1 \ v_2 \ \dots \ v_n], \quad x = (a_1, \dots, a_n)^T$$

The columns of A are the given vectors. The system $Ax = v$ has a solution $\Leftrightarrow v$ lies in the column space of A .

2.4 Existence and Uniqueness

Existence Theorem: v is a linear combination of $\{v_1, \dots, v_n\}$ if and only if $\text{rank}([A|v]) = \text{rank}(A)$, i.e., the augmented system $Ax = v$ is consistent.

Uniqueness Theorem: If $\{v_1, \dots, v_n\}$ is linearly independent, then every vector in their span has a UNIQUE representation as a linear combination. Proof: suppose $v = \sum a_i v_i = \sum b_i v_i \rightarrow \sum (a_i - b_i) v_i = 0 \rightarrow$ all $a_i = b_i$ by independence.

Condition	Consistency?	Representation unique?
$\text{rank}(A) = \text{rank}([A v])$	YES — consistent	Only if vectors are independent
$\text{rank}(A) \neq \text{rank}([A v])$	NO — inconsistent	N/A — not a linear combination
Vectors linearly independent	If consistent	YES — unique
Vectors linearly dependent	If consistent	NO — infinitely many representations

2.5 Step-by-Step Solution Method

- Write $v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$.
- Expand and equate corresponding components to form a linear system in $c_1, \dots, c_n$.
- Write the augmented matrix $[A|v]$ where $A = [v_1 | v_2 | \dots | v_n]$.
- Row-reduce $[A|v]$ to REF or RREF.
- If $\text{rank}(A) \neq \text{rank}([A|v])$: system inconsistent $\rightarrow v$ is NOT a linear combination.
- If $\text{rank}(A) = \text{rank}([A|v])$: solve for c_i and write the combination explicitly.

2.6 Worked Examples — Vectors in $\mathbb{R}^n$

Example 1 — Unique Representation

Express $(3, 5, 2)$ as a linear combination of $v_1 = (1, 1, 0)$, $v_2 = (2, 3, 0)$, $v_3 = (0, 0, 1)$.

Set up: $(3, 5, 2) = c_1(1, 1, 0) + c_2(2, 3, 0) + c_3(0, 0, 1)$. Equating components:

$$c_1 + 2c_2 = 3$$

$$c_1 + 3c_2 = 5$$

$$c_3 = 2$$

From equations 1 & 2: $c_2 = 2$, $c_1 = -1$. From eq. 3: $c_3 = 2$.

$$\text{Answer: } (3,5,2) = -1 \cdot (1,1,0) + 2 \cdot (2,3,0) + 2 \cdot (0,0,1)$$

Example 2 — Infinitely Many Representations

Express $(2, -1, -8)$ as a linear combination of $(1, 2, 1)$, $(1, 1, -1)$, $(4, 5, -2)$.

Augmented matrix after row reduction:

$$[1 \ 1 \ 4 \mid 2]$$

$$[0 \ 1 \ 3 \mid 5]$$

$$[0 \ 0 \ 0 \mid 0]$$

Consistent (rank 2 = rank 2). Free variable: $c_3 = k$. Then $c_2 = 5 - 3k$, $c_1 = -3 - k$.

$$\text{Answer: } (2, -1, -8) = (-3 - k)(1, 2, 1) + (5 - 3k)(1, 1, -1) + k(4, 5, -2) \text{ for any } k \in \mathbb{R}$$

Example 3 — NOT a Linear Combination

Is $(2, -5, 3)$ a linear combination of $(1, -3, 2)$, $(2, -4, -1)$, $(1, -5, 7)$?

Row-reduce the augmented matrix:

$$[1 \ 2 \ 1 \mid 2]$$

$$[0 \ 2 \ -2 \mid 1]$$

$$[0 \ 0 \ 0 \mid 3] \leftarrow \text{last row: } 0 = 3, \text{ contradiction!}$$

$\text{rank}([A|v]) = 3 \neq \text{rank}(A) = 2$. System inconsistent.

$$\text{Answer: } (2, -5, 3) \text{ is NOT a linear combination of the given vectors.}$$

Example 4 — Linear Combination of Matrices

Express $A = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ as a linear combination of $M_1 = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$, $M_2 = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$, $M_3 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.

Equate entry by entry: $c_1 + c_2 + c_3 = 3$, $-c_1 + c_2 + c_3 = -1$, $-c_2 = 1$, $-c_1 = -2$.

Solving: $c_1 = 2$, $c_2 = -1$, $c_3 = 2$.

$$\text{Answer: } A = 2M_1 - M_2 + 2M_3$$

2.7 Applications in Engineering

Field	Use of Linear Combinations
Signal Processing	Any signal = linear combination of basis signals (Fourier decomposition)
Structural Mechanics	Force vectors combined to find resultant or decompose into components
Electrical Circuits	Superposition: total response = sum of individual source responses
Machine Learning	Feature vector = weighted sum of basis features
Quantum Mechanics	Quantum state = superposition (linear combination) of eigenstates

Chapter 3: Linear Independence and Dependence

3.1 Definition

Linearly Independent: Vectors $\{v_1, \dots, v_n\} \subseteq V$ are linearly independent if: $a_1v_1 + \dots + a_nv_n = \mathbf{0}$ implies $a_1 = a_2 = \dots = a_n = 0$ (only trivial solution to the homogeneous equation).

Linearly Dependent: Vectors $\{v_1, \dots, v_n\}$ are linearly dependent if there exist scalars $a_1, \dots, a_n$ NOT ALL ZERO such that $a_1v_1 + \dots + a_nv_n = \mathbf{0}$. Equivalently, at least one vector is a linear combination of the others.

3.2 Equivalent Characterizations

Statement	Independent	Dependent
Homogeneous system $a_1v_1 + \dots = \mathbf{0}$	Only trivial solution	Non-trivial solution exists
One vector expressible from others	No vector = comb. of others	At least one vector redundant
Matrix rank ($A = [v_1 \dots v_n]$)	$\text{rank}(A) = n$ (# of columns)	$\text{rank}(A) < n$
Determinant (square matrix only)	$\det(A) \neq 0$	$\det(A) = 0$
Span after removal	Span shrinks	Span unchanged

3.3 Testing Methods

Method 1 — Determinant Test (n vectors in $\mathbb{R}^n$ only)

15. Form $n \times n$ matrix A with vectors as columns.
16. Compute $\det(A)$.
17. $\det(A) \neq 0 \rightarrow$ independent; $\det(A) = 0 \rightarrow$ dependent.

(If you are confused about the numbering - it is not a mistake. Instead of repeating the steps everywhere, only new ones have been mentioned. Refer above for previous steps)

Method 2 — Rank Criterion (General)

18. Form matrix $A = [v_1 \mid v_2 \mid \cdots \mid v_n]$.
19. Row-reduce A to REF.
20. Count pivots = $\text{rank}(A)$.
21. $\text{rank}(A) = n \rightarrow$ independent; $\text{rank}(A) < n \rightarrow$ dependent.

Rank Criterion: Vectors $v_1, \dots, v_n$ are linearly independent if and only if $\text{rank}([v_1 \mid \cdots \mid v_n]) = n$.
 Proof: independence $\Leftrightarrow Ax=0$ has only trivial solution $\Leftrightarrow \text{nullity}(A)=0 \Leftrightarrow \text{rank}=n$ (by Rank-Nullity).

Method 3 — Inspection

- If any vector = scalar multiple of another $\rightarrow$ dependent.
- If any vector = $0 \rightarrow$ set is dependent.
- More than n vectors in $\mathbb{R}^n \rightarrow$ always dependent ($\text{rank} \leq n$).

3.4 Fundamental Theorems

Theorem 3.1: Any set containing the zero vector is linearly dependent. Proof: $1 \cdot 0 + 0 \cdot v_2 + \cdots = 0$ is a non-trivial solution.

Theorem 3.2: Any set of more than n vectors in $\mathbb{R}^n$ is linearly dependent. Proof: A has shape $n \times m$ with $m > n$, so $\text{rank}(A) \leq n < m$, hence $\text{nullity} > 0$, so $Ax=0$ has non-trivial solutions.

Theorem 3.3: Every subset of a linearly independent set is linearly independent. Proof: removing vectors cannot introduce a non-trivial dependence.

3.5 Geometric Interpretation

Space	Independent Condition	Geometric Picture
$\mathbb{R}^2$	2 vectors, not collinear	Two distinct directions (not on same line)
$\mathbb{R}^2$	1 vector, non-zero	Single direction — spans a line
$\mathbb{R}^3$	3 vectors, no two collinear & no three coplanar	Three distinct 3D directions
$\mathbb{R}^3$	2 vectors, non-collinear	Span a plane
$\mathbb{R}^n$ general	rank = number of vectors	Each adds a new dimension

3.6 Worked Examples

Example 1 — Dependent (rank check)

Test: $(1,2,3), (2,4,6), (1,1,1)$.

Note: $(2,4,6) = 2 \cdot (1,2,3)$. Dependent by inspection.

Verify: A row-reduces to rank $2 < 3$. Dependent.

Example 2 — Independent ($\det \neq 0$)

Test: $(1,0,1), (0,1,1), (1,1,0)$.

$$\det(A) = 1(0-1) - 0 + 1(0-1) = -1-1 = -2 \neq 0$$

Result: Linearly independent.

Example 3 — Polynomial Space

Test independence of $\{1+x, x+x^2, 1+x^2\}$ in P_2 .

Suppose $a(1+x) + b(x+x^2) + c(1+x^2) = 0$. Collect by degree:

$$\text{constant: } a + c = 0$$

$$\text{degree 1: } a + b = 0$$

$$\text{degree 2: } b + c = 0$$

From these: $a = -c, b = -a = c, b + c = 2c = 0 \rightarrow c = 0 \rightarrow a = b = 0$. Only trivial solution.

Result: $\{1+x, x+x^2, 1+x^2\}$ is linearly independent.

Example 4 — Function Space

Test: $\{e^x, e^{2x}, e^{3x}\}$.

Assume $ae^x + be^{2x} + ce^{3x} = 0$ for all x . Divide by e^x : $a + be^x + ce^{2x} = 0$. Differentiate twice and evaluate — only trivial solution exists.

Result: $\{e^x, e^{2x}, e^{3x}\}$ is linearly independent.

3.7 Common Errors

Error	Correction
Assuming orthogonal $\Rightarrow$ independent	Orthogonality implies independence, but independence does NOT require orthogonality
Testing only a subset of equations	Must verify the FULL homogeneous system, not just pairs
Using det for non-square matrices	Determinant test only valid for n vectors in $\mathbb{R}^n$; use rank for general case
Confusing dependent with parallel	Dependence is more general than parallelism (applies in all dimensions)

Chapter 4: Span of a Set of Vectors

4.1 Definition

Span: Let V be a vector space and $S = \{v_1, \dots, v_n\} \subseteq V$. The span of S is the set of ALL linear combinations of the vectors in S :

$$\text{Span}\{v_1, \dots, v_n\} = \{a_1 v_1 + \dots + a_n v_n \mid a_1, \dots, a_n \in F\}$$

The span is the smallest subspace containing S .

4.2 Span as a Subspace

Theorem 4.1: For any $S \subseteq V$, $\text{Span}(S)$ is a subspace of V . Proof: (i) $0 = 0 \cdot v_1 + \dots + 0 \cdot v_n \in \text{Span}(S)$. (ii) If $u, w \in \text{Span}(S)$ then $u+w = \sum(a_i+b_i)v_i \in \text{Span}(S)$. (iii) $cu = \sum(ca_i)v_i \in \text{Span}(S)$.

Smallest Subspace Property: $\text{Span}(S)$ is the smallest subspace of V containing S . Any subspace W that contains S must satisfy $\text{Span}(S) \subseteq W$.

4.3 Matrix Interpretation

If $A = [v_1 \mid v_2 \mid \dots \mid v_n]$, then:

$$\text{Span}\{v_1, \dots, v_n\} = \text{Col}(A) = \{Ax \mid x \in \mathbb{R}^n\}$$

Testing if vector $b \in \text{Span}\{v_1, \dots, v_n\} \Leftrightarrow$ testing if $Ax = b$ is consistent.

4.4 Geometric Pictures

Vectors	Independence?	Span in $\mathbb{R}^2$	Span in $\mathbb{R}^3$
1 non-zero vector	N/A	Line through origin	Line through origin
2 vectors	Dependent (collinear)	Line through origin	Line through origin
2 vectors	Independent	All of $\mathbb{R}^2$	Plane through origin
3 vectors	Independent	All of $\mathbb{R}^2$	All of $\mathbb{R}^3$
3 vectors	Dependent	Line or plane	Plane or line

4.5 Generating Sets

Generating Set / Spanning Set: $S \subseteq V$ is a generating set (spanning set) for V if $\text{Span}(S) = V$. i.e., every vector in V can be expressed as a linear combination of vectors in S .

Minimal Spanning Set: A spanning set S such that no vector can be removed while preserving $\text{Span}(S) = V$. Equivalent to: S spans V AND S is linearly independent, i.e., S is a BASIS.

Theorem 4.2: A spanning set is minimal $\Leftrightarrow$ it is linearly independent. If dependent, remove a redundant vector; span is unchanged. If independent, removing any vector shrinks the span.

4.6 Worked Examples

Example 1 — Span equals entire space

$\text{Span}\{(1,0), (0,1)\}$ in $\mathbb{R}^2$. For any (a,b) : $(a,b) = a(1,0) + b(0,1)$. Hence $\text{Span} = \mathbb{R}^2$.

Example 2 — Span is a line

$\text{Span}\{(2,4)\}$ in $\mathbb{R}^2$. All scalar multiples: $(2a, 4a)$. This is the line $y = 2x$.

Example 3 — Membership test

Does $(3,5) \in \text{Span}\{(1,2), (2,1)\}$? Solve: $a(1,2) + b(2,1) = (3,5)$.

$$a + 2b = 3$$

$$2a + b = 5$$

Adding: $3a + 3b = 8$ — subtract 1st from 2nd: $a - b = 2$. Solve: $a = 13/3$, $b = 7/3$... let's re-check: $a + 2b = 3$ and $2a + b = 5$. Multiply eq1 by 2: $2a + 4b = 6$. Subtract eq2: $3b = 1$, $b = 1/3$, $a = 7/3$.

Solution exists, so $(3,5) \in \text{Span}$.

Example 4 — Span in polynomial space

$\text{Span}\{1+x, x+x^2, 1+x^2\}$: any $p(x) = a+bx+cx^2 = \alpha(1+x) + \beta(x+x^2) + \gamma(1+x^2)$. Solve for α, β, γ .

System: $\alpha + \gamma = a$, $\alpha + \beta = b$, $\beta + \gamma = c$. Unique solution exists ($\det \neq 0$). Hence $\text{Span} = P_2$.

4.7 Span of Infinite Sets

For infinite sets S , $\text{Span}(S)$ = all FINITE linear combinations. No infinite sums.

Example: $\text{Span}\{1, x, x^2, x^3, \dots\}$ = all polynomials of finite degree = P (polynomial space).

Chapter 5: Basis and Dimension

5.1 Basis — Definition

Basis: A set $B = \{v_1, \dots, v_n\} \subseteq V$ is a basis of V if: (1) B SPANS V , i.e., $\text{Span}(B) = V$, AND (2) B is LINEARLY INDEPENDENT. Equivalently: B is a minimal spanning set, or B is a maximal independent set.

5.2 Dimension — Definition

Dimension: If V has a finite basis of n vectors, then $\dim(V) = n$. V is called finite-dimensional. If no finite basis exists, V is infinite-dimensional.

Invariance Theorem (Theorem 5.1): All bases of a finite-dimensional vector space V contain the same number of vectors. Proof: Let B_1, B_2 be bases. Since B_1 independent and B_2 spans: $|B_1| \leq |B_2|$. Symmetrically $|B_2| \leq |B_1|$. Hence $|B_1| = |B_2|$. This justifies the definition of dimension.

5.3 Standard Bases and Dimensions

Space	Standard Basis	Dimension
$\mathbb{R}^n$	$e_1=(1,0,\dots,0), e_2=(0,1,\dots,0), \dots,$ $e_n=(0,\dots,0,1)$	n
P_n (polynomials degree $\leq n$)	$\{1, x, x^2, \dots, x^n\}$	$n+1$
$M_{m \times n}$ ($m \times n$ matrices)	$\{E_{ij} : \text{each has 1 in position } (i,j), 0 \text{ elsewhere}\}$	$m \cdot n$
$\mathbb{R}$ (scalars)	$\{1\}$	1
$\{0\}$ (trivial space)	Empty set $\emptyset$	0

5.4 Basis Theorems

Theorem 5.2 — Basis Characterization: Let $\dim(V) = n$. A set of exactly n vectors is a basis $\Leftrightarrow$ it is (a) linearly independent, OR (b) spans V . (Any one condition implies the other for sets of exactly n vectors.)

Theorem 5.3 — Dimension Bounds: In a space with $\dim(V) = n$: (a) Any spanning set has $\geq n$ vectors. (b) Any independent set has $\leq n$ vectors.

Theorem 5.4 — Subspace Dimension: If W is a subspace of V , then $\dim(W) \leq \dim(V)$. If W is a PROPER subspace, then $\dim(W) < \dim(V)$.

Theorem 5.5 — Extension Theorem: Every linearly independent set in a finite-dimensional V can be extended to a basis of V . Every spanning set can be reduced to a basis (by removing dependent vectors).

5.5 Matrix Interpretation

Matrix Concept	Vector Space Concept
$\text{Rank}(A)$ = number of pivots in RREF	$\dim(\text{Col}(A)) = \dim(\text{column space})$
$\text{Nullity}(A)$ = number of free variables	$\dim(N(A)) = \dim(\text{null space})$
Pivot columns of A	Basis for $\text{Col}(A)$
Non-zero rows of $\text{RREF}(A)$	Basis for $\text{Row}(A)$
$\text{rank}(A) + \text{nullity}(A) = n$ (columns)	Rank-Nullity Theorem

5.6 Basis Extraction Procedure

22. Form matrix A with candidate vectors as columns.
23. Row-reduce A to RREF.
24. Identify pivot positions (columns with leading 1s).
25. The ORIGINAL (pre-reduction) columns of A at pivot positions form a basis for $\text{Col}(A)$.

WARNING: Use the original columns of A , NOT the columns of $\text{RREF}(A)$. Row operations change the column space, not the row space.

5.7 Gram–Schmidt Orthogonalization

Given independent set $\{v_1, \dots, v_n\}$, produce orthogonal basis $\{u_1, \dots, u_n\}$ spanning same subspace:

$$u_1 = v_1$$

$$u_2 = v_2 - \text{proj}_{u_1}(v_2)$$

$$u_3 = v_3 - \text{proj}_{u_1}(v_3) - \text{proj}_{u_2}(v_3)$$

$$u_i = v_i - \sum_{j=1}^{i-1} \text{proj}_{u_j}(v_i)$$

Where the projection formula is:

$$\text{proj}_u(v) = (\langle v, u \rangle / \langle u, u \rangle) \cdot u$$

Normalize to get orthonormal basis: $\hat{e}_i = u_i / \|u_i\|$

5.8 Change of Basis

Let $B = \{v_1, \dots, v_n\}$ and $C = \{w_1, \dots, w_n\}$ be two bases of V . The change-of-basis matrix P from C to B has columns = coordinate vectors of w_i relative to B :

$$[v]_B = P \cdot [v]_C$$

$$[v]_C = P^{-1} \cdot [v]_B$$

P is always invertible (since both sets are bases).

5.9 Worked Examples

Example 1 — Find $\dim(W)$ from constraints

$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. One constraint $\rightarrow$ 2 free variables ($y=s, z=t$):

$$(x, y, z) = s(-1, 1, 0) + t(-1, 0, 1)$$

Basis: $\{(-1, 1, 0), (-1, 0, 1)\}$, $\dim(W) = 2$

Example 2 — Dimension by inspection

$W = \text{Span}\{(1, 2), (2, 4), (3, 6)\}$. All multiples of $(1, 2)$.

$\dim(W) = 1$, Basis = $\{(1, 2)\}$

Example 3 — Gram-Schmidt

Apply G-S to $\{(1, 1, 0), (1, 0, 1)\}$:

$$u_1 = (1, 1, 0)$$

$$\text{proj}_{u_1}(v_2) = \langle (1, 0, 1), (1, 1, 0) \rangle / \langle (1, 1, 0), (1, 1, 0) \rangle \cdot (1, 1, 0)$$

$$= (1/2)(1, 1, 0) = (1/2, 1/2, 0)$$

$$u_2 = (1, 0, 1) - (1/2, 1/2, 0) = (1/2, -1/2, 1)$$

Orthogonal basis: $\{(1, 1, 0), (1/2, -1/2, 1)\}$

Chapter 6: Subspaces, Null Space, Column Space, and Rank

6.1 Subspaces

Subspace: A nonempty subset $W \subseteq V$ is a subspace of V if it is itself a vector space under the same operations. Equivalently (Subspace Test): $u, v \in W$ and $c \in F \Rightarrow u + cv \in W$. (This single condition implies all vector space axioms given W is nonempty.)

Standard Examples of Subspaces

Subspace	Definition	Is a Subspace?
Span of vectors	$\text{Span}\{v_1, \dots, v_n\}$	YES — closed under lin. comb.
Null space $N(A)$	$\{x : Ax=0\}$	YES — proved below
Column space $\text{Col}(A)$	Span of columns of A	YES — is a span
Row space $\text{Row}(A)$	Span of rows of A	YES — is a span
Line not through origin	$\{(1,2)+t(3,4) : t \in \mathbb{R}\}$	NO — doesn't contain 0
$\{0\}$	Zero set	YES — trivial subspace

6.2 The Four Fundamental Subspaces

For matrix $A \in \mathbb{R}^{m \times n}$:

Subspace	Lives in	Dimension	How to find basis
Column Space $\text{Col}(A)$	$\mathbb{R}^m$	$\text{rank}(A)$	Original pivot columns of A
Row Space $\text{Row}(A)$	$\mathbb{R}^n$	$\text{rank}(A)$	Non-zero rows of $\text{RREF}(A)$
Null Space $N(A)$	$\mathbb{R}^n$	$n - \text{rank}(A) = \text{nullity}(A)$	Solve $Ax=0$; parameterize free vars
Left Null Space $N(A^T)$	$\mathbb{R}^m$	$m - \text{rank}(A)$	Solve $A^T y=0$

Orthogonality of Fundamental Subspaces: $\text{Row}(A) \perp N(A)$ in $\mathbb{R}^n$, and $\text{Col}(A) \perp N(A^T)$ in $\mathbb{R}^m$. These are orthogonal complements: $\dim(\text{Row}(A)) + \dim(N(A)) = n$.

6.3 Null Space (Kernel)

Null Space: $N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$. The null space is the solution set of the homogeneous system. It is always a subspace of $\mathbb{R}^n$.

Proof $N(A)$ is a subspace: If $Ax=0$ and $Ay=0$, then $A(x+y) = Ax+Ay = 0+0 = 0$, and $A(cx) = c(Ax) = 0$. Both remain in $N(A)$.

Algorithm: Finding a Basis for $N(A)$

26. Row-reduce A to RREF.
27. Identify free variables (non-pivot columns).
28. For each free variable x_i , set $x_i=1$ and all other free variables=0.
29. Back-substitute to express pivot variables.
30. Each resulting vector is a basis vector for $N(A)$.

6.4 Column Space

Column Space: $\text{Col}(A) = \text{Span}\{a_1, a_2, \dots, a_n\} = \{Ax : x \in \mathbb{R}^n\}$. The column space is the set of all possible outputs of the linear transformation $x \mapsto Ax$.

Consistency Criterion: The system $Ax = b$ is consistent $\Leftrightarrow b \in \text{Col}(A) \Leftrightarrow \text{rank}([A|b]) = \text{rank}(A)$.

Algorithm: Finding a Basis for $\text{Col}(A)$

31. Row-reduce A to RREF.
32. Identify pivot columns in RREF.
33. Take the CORRESPONDING ORIGINAL columns of A (before reduction).

Critical Distinction: Basis for $\text{Col}(A)$ uses original columns at pivot positions. Basis for $\text{Row}(A)$ uses the non-zero rows of RREF directly.

6.5 Rank of a Matrix

Rank: $\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A)) = \text{number of pivot positions in RREF}(A)$.

Property	Statement
Rank equals row rank	$\dim(\text{Col}(A)) = \dim(\text{Row}(A))$ always
Transpose rank	$\text{rank}(A) = \text{rank}(A^T)$
Rank bound	$\text{rank}(A) \leq \min(m,n)$ for $A \in \mathbb{R}^{m \times n}$
Product rank bound	$\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
Full column rank	$\text{rank}(A)=n \Leftrightarrow Ax=0$ has only trivial solution $\Leftrightarrow$ columns independent
Full row rank	$\text{rank}(A)=m \Leftrightarrow Ax=b$ consistent for every $b \in \mathbb{R}^m$

6.6 Worked Examples

Example 1 — Finding Null Space

Find $N(A)$ where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$.

RREF: $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

Free vars: $x_2=s, x_3=t$. From row 1: $x_1 = -2s-3t$.

$N(A)$ basis: $\{(-2, 1, 0), (-3, 0, 1)\}$. $\dim(N(A)) = 2$.

Example 2 — Rank and Col(A)

$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 1 \end{bmatrix}$. Row reduce:

$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Pivots in columns 1 and 3. Rank = 2.

Basis for $\text{Col}(A)$: $\{\text{col}_1=(1,2,3), \text{col}_3=(1,1,1)\}$ — use original columns at pivot positions.

Example 3 — Consistency Check

Is $Ax=b$ consistent with $A=[[1,1],[2,2]]$, $b=(3,6)$?

$$[A|b] = [1 \ 1 \ | \ 3] \rightarrow [1 \ 1 \ | \ 3]$$

$$[2 \ 2 \ | \ 6] \quad [0 \ 0 \ | \ 0]$$

$\text{rank}(A) = \text{rank}([A|b]) = 1$. Consistent.

$b \in \text{Col}(A)$: YES. Solution: $x_1 = 3 - x_2$ for any x_2 .

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Chapter 7: The Rank–Nullity Theorem

7.1 Statement of the Theorem

Rank-Nullity Theorem: For any matrix $A \in \mathbb{R}^{m \times n}$: $\text{rank}(A) + \text{nullity}(A) = n$ where n = number of COLUMNS. Equivalently, for linear transformation $T: V \rightarrow W$: $\dim(\ker T) + \dim(\text{Im } T) = \dim(V)$

$$\text{rank}(A) + \text{nullity}(A) = n$$

7.2 Definitions

Term	Definition	Equals
$\text{rank}(A)$	$\dim(\text{Col}(A))$	Number of pivots in $\text{RREF}(A)$
$\text{nullity}(A)$	$\dim(N(A))$	Number of free variables in $Ax=0$
$\ker(T)$	Kernel of $T = \{v : T(v)=0\}$	Null space for matrix representation
$\text{Im}(T)$	Image of $T = \{T(v) : v \in V\}$	Column space for matrix representation

7.3 Interpretation

Think of the n variables in a system as being divided into two groups:

Variable Type	Count	Connection
Leading (pivot) variables	$\text{rank}(A) = r$	Dimension of image / column space
Free variables	$\text{nullity}(A) = n - r$	Dimension of kernel / null space
Total variables	n	Dimension of domain

Intuition: rank counts how many independent constraints the matrix imposes; nullity counts how many degrees of freedom survive after those constraints.

7.4 Proof by Row Reduction

Let $\text{RREF}(A)$ have r pivot columns. Then:

- $r = \text{rank}(A)$ (number of pivot positions)
- $n-r = \text{nullity}(A)$ (number of free variable columns)
- Null space has dimension $n-r$ (one basis vector per free variable)

Therefore: $\text{rank}(A) + \text{nullity}(A) = r + (n-r) = n$. $\square$

7.5 Proof via Linear Transformations

Let $T: V \rightarrow W$. Let $\{k_1, \dots, k_m\}$ be a basis for $\ker(T)$. Extend to a basis of V :

$\{k_1, \dots, k_m, v_{m+1}, \dots, v_n\}$.

Claim: $\{T(v_{m+1}), \dots, T(v_n)\}$ is a basis for $\text{Im}(T)$.

- Spanning: any $T(v)$ can be written in terms of these — the $\ker$ -part maps to 0 .
- Independence: if $\sum c_i T(v_{m+i}) = 0$, then $T(\sum c_i v_{m+i}) = 0$, so $\sum c_i v_{m+i} \in \ker(T)$ — linear combination of k 's, but also of v_{m+i} 's — independence forces all $c_i = 0$.

Hence $\dim(\ker T) = m$, $\dim(\text{Im } T) = n-m$, and their sum $= n = \dim(V)$. $\square$

7.6 Geometric Cases in $\mathbb{R}^3$

rank(A)	nullity(A)	Geometric Description	Example
3	0	Bijection: $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ (invertible)	Rotation/scaling in 3D
2	1	Collapses 3D to plane; kernel = line	Projection onto a plane
1	2	Collapses 3D to line; kernel = plane	Projection onto a line
0	3	Zero map: everything $\rightarrow 0$	All-zeros matrix

7.7 Relation to Solution Structure of $Ax = b$

Solution Structure: If $Ax = b$ is consistent, its complete solution is: $x = x_{\square} + x_{\square}$, where $x_{\square}$ is any particular solution ($Ax_{\square} = b$) and $x_{\square} \in N(A)$ is the general homogeneous solution. The solution set is an affine subspace of dimension = nullity(A).

nullity(A)	rank(A)=n?	Solution of $Ax=b$ (if consistent)
0	YES (full col. rank)	Unique solution
1	NO	1-parameter family of solutions
2	NO	2-parameter family of solutions
k	NO	k-parameter family of solutions

7.8 Worked Examples

Example 1 — Direct Application

A is 4×6 matrix with rank(A) = 3. Find nullity.

$$3 + \text{nullity} = 6 \Rightarrow \text{nullity} = 3$$

Example 2 — Full Computation

Find rank and nullity of $A = [[1,2,3],[2,4,6],[1,1,1]]$.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 5 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 5 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

After full RREF: 2 pivots. rank(A) = 2. nullity = $3 - 2 = 1$.

Ans: rank = 2, nullity = 1, and $2 + 1 = 3 = n$ ✓

Example 3 — Large System

A is 5×8 with rank 4. Find dimension of solution space of $Ax=0$.

$$\text{nullity} = 8 - 4 = 4$$

Solution space has dimension 4 (four free parameters).

Example 4 — Linear Transformation

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x,y,z) = (x+y, y+z)$. Matrix:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Row reduce: rank = 2. Nullity = $3-2 = 1$.

$\ker(T)$: Solve $Ax=0$: $x = t(1, -1, 1)$. $\dim(\ker T)=1$, $\dim(\text{Im } T)=2$. $1+2=3=\dim(\mathbb{R}^3)$ ✓

Example 5 — Derivative Transformation

$T: P_3 \rightarrow P_2$, $T(p) = p'$ (differentiation). $\dim(P_3) = 4$, $\dim(P_2) = 3$.

$\ker(T) = \text{constant polynomials} = \text{Span}\{1\}$. $\dim(\ker T) = 1$.

$\text{Im}(T) = \text{all polynomials of degree} \leq 2$. $\dim(\text{Im } T) = 3$. Check: $1+3 = 4$ ✓

7.9 Common Errors

Error	Correction
Using m (rows) instead of n (columns) in 'rank+nullity=?'	Theorem states rank+nullity = n = NUMBER OF COLUMNS always
Confusing rank with number of rows/equations	rank = # of PIVOTS, not # of equations. Some equations may be redundant
Applying theorem to infinitely-dimensional spaces	Classical theorem requires finite-dimensional domain
Forgetting to check full rank for uniqueness	Unique solution to $Ax=b$ iff $\text{rank}(A)=n$ (not just consistency)

Formula Sheet — Module 2 Quick Reference

Homogeneous Systems

Formula / Fact	Condition / Notes
$Ax = 0$ always consistent	Trivial solution $x=0$ always exists
$\det(A) \neq 0 \Leftrightarrow$ only trivial solution	Square $n \times n$ matrix
$\det(A) = 0 \Leftrightarrow$ non-trivial solutions	Square $n \times n$ matrix
$\dim(N(A)) = n - \text{rank}(A)$	Rank-Nullity applied to null space
Solution set = null space $N(A)$	Always a subspace of $\mathbb{R}^n$

Linear Combinations

Formula / Fact	Condition / Notes
$v = a_1 v_1 + \dots + a_n v_n$	Definition; a_i are coefficients
$v \in \text{Span}\{v_i\} \Leftrightarrow Ax=v$ consistent	$A = [v_1 \dots v_n]$
Unique repr. $\Leftrightarrow \{v_i\}$ independent	Uniqueness theorem
$\text{rank}(A) = \text{rank}([A v]) \Leftrightarrow$ consistent	Rouché–Capelli condition

Linear Independence

Formula / Fact	Condition / Notes
$a_1 v_1 + \dots + a_n v_n = 0 \Rightarrow$ all $a_i = 0$	Definition of independence
$\text{rank}([v_1 \dots v_n]) = n \Leftrightarrow$ independent	Rank criterion
$\det(A) \neq 0 \Leftrightarrow$ independent	Square matrix only

Formula / Fact	Condition / Notes
More than n vectors in $\mathbb{R}^n \Rightarrow$ dependent	Rank $\leq n$
Set containing $0 \Rightarrow$ dependent	Trivially: $1 \cdot 0 = 0$ is non-trivial

Span

Formula / Fact	Condition / Notes
$\text{Span}\{v_1, \dots, v_n\} = \{\sum a_i v_i : a_i \in F\}$	All linear combinations
$\text{Span}(S) = \text{Col}(A), A = [v_1 \dots v_n]$	Matrix interpretation
$\text{Span}(S)$ is always a subspace	Closed under $+$ and $\cdot$
$\text{Span}(S) = V \Leftrightarrow S$ is a spanning set	Generating set definition
Minimal span $\Leftrightarrow$ independent	Minimal spanning set = basis

Basis and Dimension

Formula / Fact	Condition / Notes
Basis = spanning + independent	Both conditions required
$\dim(\mathbb{R}^n) = n$	Standard
$\dim(P_n) = n+1$	Polynomials of degree $\leq n$
$\dim(M_{m \times n}) = mn$	Space of $m \times n$ matrices
n vectors in n -dim V : independent $\Leftrightarrow$ basis $\Leftrightarrow$ spanning	Characterization theorem
W proper subspace $\Rightarrow \dim(W) < \dim(V)$	Strict inequality

Rank-Nullity Theorem

Formula / Fact	Condition / Notes
$\text{rank}(A) + \text{nullity}(A) = n$	A is $m \times n$; $n = \#$ of COLUMNS
$\dim(\ker T) + \dim(\text{Im } T) = \dim(V)$	$T: V \rightarrow W$
$\text{rank}(A) = \text{rank}(A^T)$	Row rank = column rank
$\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$	Product inequality
$A \text{ invertible} \Leftrightarrow \text{rank}(A)=n \Leftrightarrow \text{nullity}(A)=0$	$n \times n$ square matrix
Solution space of $Ax=b$ has $\dim = \text{nullity}(A)$	When consistent

Subspaces and Fundamental Spaces

Subspace	Lives in	dim	Basis from
$\text{Col}(A)$	$\mathbb{R}^m$	$\text{rank}(A)$	Pivot columns of original A
$\text{Row}(A)$	$\mathbb{R}^n$	$\text{rank}(A)$	Non-zero rows of RREF(A)
$N(A)$	$\mathbb{R}^n$	$n - \text{rank}(A)$	Free-variable vectors from $Ax=0$
$N(A^T)$	$\mathbb{R}^m$	$m - \text{rank}(A)$	Free-variable vectors from $A^T y=0$

Exam Quick Reference Card

Concept	Test	Result if TRUE
Homogeneous: only trivial sol.?	$\det(A) \neq 0$ OR $\text{rank}(A)=n$	$x=0$ is only solution
v is lin. combination of $\{v_i\}$?	$\text{rank}(A) = \text{rank}([A v])$	YES, system consistent
$\{v_i\}$ linearly independent?	$\text{rank}([v_1 \dots v_n]) = n$	YES, independent

Concept	Test	Result if TRUE
S spans V?	$\text{Span}(S)=V$ (check dim of span)	S is a spanning set
B is a basis?	Independent AND spans	$\dim(V)= B $
$Ax=b$ unique solution?	$\text{rank}(A)=n$	Unique solution
$Ax=b$ consistent?	$\text{rank}(A)=\text{rank}([A b])$	Solution(s) exist

Practice Problem Bank — CIE 1 Preparation

Section A: Homogeneous Systems

- Q1. Solve the homogeneous system: $x+2y+3z=0$, $2x+4y+6z=0$, $x-y+z=0$.
- Q2. For what value of k does $kx+y-z=0$, $x+ky-z=0$, $x+y-kz=0$ have a non-trivial solution?
- Q3. Find a basis for the null space of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 0 & 1 & 2 & 3 \end{bmatrix}$.

Section B: Linear Combinations

- Q4. Express $(5, -2, 4)$ as a linear combination of $(1, 2, 0)$, $(2, 1, 1)$, $(1, -1, 2)$.
- Q5. Verify if $(3, -1, 5)$ is a linear combination of $(1, -2, 1)$, $(2, 1, 3)$, $(1, 0, 2)$.
- Q6. Show that $A = \begin{bmatrix} 4 & 2 \\ 1 & -3 \end{bmatrix}$ is a linear combination of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Section C: Linear Independence

- Q7. Test the independence of $(2, 1, 0)$, $(1, 1, 1)$, $(0, 1, 2)$. Find the dependence relation if dependent.
- Q8. Are $\{1+x, x+x^2, 1+x^2-x\}$ independent in P_2 ?
- Q9. Prove or disprove: $\{\sin^2 x, \cos^2 x, \cos 2x\}$ are linearly dependent over $\mathbb{R}$.

Section D: Span, Basis, Dimension

- Q10. Find a basis and dimension for $W = \{(x, y, z, w) \in \mathbb{R}^4 : x+y=0, z+w=0\}$.
- Q11. Extend $\{(1, 0, 1), (0, 1, 1)\}$ to a basis of $\mathbb{R}^3$.
- Q12. Apply Gram-Schmidt to $\{(1, 1, 1), (1, 0, 1), (0, 1, 1)\}$.

Section E: Rank-Nullity

Q13. A is a 6×9 matrix with rank 4. Find: (a) nullity, (b) $\dim(\text{Col } A)$, (c) $\dim(N(A))$.

Q14. $T: P_4 \rightarrow P_3$ defined by $T(p) = p'$. Find rank and nullity. Verify Rank-Nullity.

Q15. For $A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 1 & 3 & 3 & 1 \end{bmatrix}$, find rank, nullity, basis for $N(A)$, basis for $\text{Col}(A)$.

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